

ID: ____

Training Condition: ____

Task Order: ____

Before the study

- ☐ Check ID, Training Condition, and Task Order from <link-to-recruitment-screening>
- ☐ Reserve room for observation OR send zoom link
 - ☐ Update calendar event with location and Zoom link (for recording)
- ☐ **Email** event confirmation and zoom link
- ☐ Copy this doc for each participant
- ☐ Make a copy of the study_2_starter folder and rename it with the PID
- ☐ Start Jupyter

```
BROWSER='/Applications/Google Chrome.app/Contents/MacOS/Google Chrome' jupyter notebook
```
- ☐ Start the logging server

```
ruby usage_events.rb
```

5 min: intro

Study overview

So you've seen Plottery before. It's a tool in Jupyter to help create matplotlib plots with code, a graphical interface, and AI assistance.

We're trying to learn how people use Plottery over a longer period of time and for different kinds of plots, like how someone might use the tool on their own. First we will practice together with the tool, then we are going to ask you to make several plots, and we'll just watch. We will leave a few minutes at the end to discuss the experience.

Any questions so far?

Have them read and fill out consent form

- ☐ Consent for study
- ☐ Consent for audio
- ☐ Consent for video

(make a copy during the first free-form task)

Study purpose and requirements

Great. The study overall shouldn't take more than 2 hours. If you need to stop the study and leave at any point, you can of course. We'll take a 5min break after about an hour.

Please keep in mind that this study is about understanding the usage of Plottery. It's not about checking your ability to write code or solve tasks. The tasks are hard and come from actual plots we've made for our papers. If you don't finish a plot, or get stuck, that's totally fine.

Besides Plottery, you can use google search or any other online resource, just like you would if you were working on your own.

Do you use tap to click? Natural scrolling? A mouse?

- ☐ Set up system prefs how they like

Screensharing and recording

I want to ask you to share your *entire* screen through Zoom and we'll be recording the session. The recording will only be available to me and a few other researchers on the team. Is that ok with you? Great, I will begin recording now.

- ☐ Start recording to cloud

~15-40mins: Practice

- ☐ Open "PRACTICE Tool Usage Preference.ipynb"
- ☐ View > Toggle Header (to hide it for more vertical space)

We will start with this practice to refamiliarize yourself with the tool. We will make this plot in two parts. First, getting all the parts onto the plot, then styling everything.

Read the task description.

For practice, for the first part you can (edit code and click around in interface but cannot use AI / edit code and prompt the AI but cannot click around in the interface). Then we'll reverse it for the second part. You can always open another tab and google. Okay, you can start, and I'll help you out.

You, the facilitator, freely offer help and walk them through the different parts of the interface.

For part 1, the plot does not have to match the picture exactly...it's just a rough cutoff point.

Alright, for part 2, now you can (edit code and prompt the AI but cannot click around in the interface / edit code and click around in interface but cannot use AI).

Again, freely offer help. The plot is complete when it is ready for a paper (as determined by the participant).

After practice:

- ☐ Ensure that events are logging

Okay! Now we'll start making other plots.

Tasks, max 40min each

Use the task order for the participant.

- ☐ Open the appropriate notebook

Okay, now you can use anything: code edits, clicking in the interface, AI, and google. I won't offer help. We'll spend up to 40 minutes on this one: don't worry if you don't finish in that time. You don't have to match the picture perfectly, but imagine you are creating plots for your paper.

Start your timer and then read the description with them to introduce the task. Then sit silently.

While you get started, I'll get you a copy of your forms.

- ☐ Copy consent for study
- ☐ Copy consent for audio
- ☐ Copy consent for video

If they get reaaaaaaaaaalllllllyyyy stuck and look to you for help, you can offer matplotlib pointers only (not how to do it in the GUI/AI).

Task is done after 40min or if they are satisfied enough they could put in their paper (submission for review or for camera ready, up to them).

After the 1hr mark (likely after first task), take a 5min break.

Okay, let's take a five minute break before the next plot. Feel free to walk around, stretch your legs, use the bathroom. We'll start the next plot in 5min.

Continue doing 40min tasks (there's no questions after each task) until there's 12min left, then cut them off: **Okay let's stop here so we have time for questions.**

- ☐ Stop the logging server, or record stop time: _____ (in case they start clicking around during the interview and produce more UI events)

10min: After Study Interview

(It's okay if you run out of time and don't get to all the questions.)

1. In Plottery, you can edit code, click in the interface, or ask the AI. When did you choose one way over the other?
2. How much did you try to understand the Matplotlib code?
3. When did the tool *fail* you?
4. When did the tool *help* you?
5. Anything else you want to say?

2 min: End

Thanks for participating! I'll stop recording now.

Here's your \$80 gift card! Please fill out this receipt.

Do you have any other questions?

Once again, thank you so much for your time.

- ☐ Stop recording
☐ Fill out gift card receipt
☐ Give gift card

Post Session Checklist

- ☐ Download and upload the recording to the g-drive
- ☐ Upload user files to g-drive
- ☐ Update <link-to-recruitment-screening> accordingly (study completion flag)